The Impact of Internet of Things Supported by Emerging 5G in Power Systems: A Review

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***Abstract*—The Internet of Things (IoT) is playing an important role in providing access to affordable, clean and green energy worldwide through the use of smart devices. The current electric power networks will be more reliable, secure, flexible and durable by implementing IoT in power systems. This paper presents a brief discussion about IoT contributions in the development of power systems from a generation, transmission, distribution and consumption point of view. 5G cellular networks have a great potential for the development of IoT technology. Additionally, 5G cellular networks can be instrumental in supporting the greater communication needs of IoT. This review provides a comprehensive analysis about the role of 5G cellular networks in the growth of IoT technology and power systems. Large amounts of data will be generated due to the incorporation of renewable energy, deployment of the smart grid and the improvements to the electricity market. In this way, in order to realize the connection between things and people, things and things and people and people in power systems, it is essential to apply IoT in power systems. In this case, 5G is providing numerous advantages to Power IoT (PIoT) by offering greater opportunities in progress and improvements; however, there are also numerous challenges with the deployment of 5G in PIoT. Finally, this review article provides an overview of the role, implications and challenges of 5G in PIoT.**

***Index Terms*—Fifth Generation (5G), Internet of Things (IoT), Power IoT (PIoT).**

I. Introduction

THE Internet of Things (IoT) is an emerging and promising technology which tends to revolutionize the global world through connected physical objects. The concept of IoT as a combination of heterogeneous smart devices, converted into real time with the arrival of the first application for automated inventory systems in 1983 [1]. However, it really took momentum, as a promising technology for the future vision of the internet, in 1999 [2]. Today, within almost any context, the growing absence of small and inexpensive computing devices, equipped with detection and communication capabilities, is paving the way for IoT perception everywhere. IoT delivers interacted connectivity to things used in daily life. All physical objects on earth (such as appliances, goods, buildings, vehicles, plants, etc.) are the "Things" in the IoT [3], [4]. By 2020, making it one of the fastest growing technologies in all computing, it is predicted that around 50 billion smart devices will be associated with IoT, which will be more than six times the world's expected population at that time [5]. The smart connection is with current networks, though IoT outcomes are pervasive, with information sensing by computers being performed without the assistance of human involvement [6]. The implementations of IoT are extended into an extensive range of applications like, homes, industries, energy systems, logistics, cities, agriculture, health care, and etc.

Electricity is a necessity and it has a great impact on our modern society and the global world. However, most of the power system architecture has been designed more than fifty years ago, and has become complex and cannot fulfill the requirements of today's modern society [7]. Importing IoT technology to power systems can effectively integrate and improve the information level, utilization of the power infrastructure, devices interaction facilities, and promote the services of advanced information and communication systems in the power grid system [8]. In power systems, communication between the large number of connected devices required the need for greater capacity, greater connectivity and higher data rates which can be achieved only by merging 5G with IoT. This is because only the 5G cellular network has promising features to provide significantly low latency, high integrity and high data rates. In addition, 5G cellular networks can play a potential role in facilitating the connectivity of a large number of devices connected to the internet [9].

In 2019, State Grid Corporation of China (SGCC), committed to becoming a world-class energy Internet enterprise with global competitiveness, announced that it will build an intelligent internet-based power network. This network will work as a hub and provide a platform to share energy information among all power systems. To achieve these objectives, the construction of an intelligent service system that combines all equipment and connections in the power system and provides an efficient communication process with humans and machines should be accelerated. This system can be employed easily and flexibly, so that all objectives can be performed and the information is processed effectively. Mobile internet recognizes the functions of this system through various advanced information technologies, from artificial intelligence (AI) to advanced communication techniques. In this way, 5G communication network technology is found to be more multi-functional and also very flexible. This provides a better platform to support many critical issues and also solve many problems with respect to the cost analysis and power applications in IoT. This review

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paper describes the IoT contributions in the development of the power system, the need for 5G technology in IoT and the areas where the power system can benefit from the improved access to data. The main objective of this paper is to demonstrate 5G cellular network technology emerging within IoT networks in the smart grid, thus bringing the IoT into the future of electrical power systems.

The rest of this paper is organized as follows. Section II provides a brief explanation of IoT development in all parts of the power system. Section III briefly describes the benefits of 5G technology emerging with IoT regarded power systems. Section IV discusses the basic concept and architecture of PIoT and also provides details about how a 5G system supports PIoT as well as some of the challenges and suggested solutions associated with the deployment of 5G in PIoT. Section V provides the conclusion of this paper.

II. Development of IoT in Controlling, Monitoring and Management of Power Systems

Electricity and energy systems are the main triggers of economic growth. Providing access to clean and inexpensive energy has become a universal priority, and countries around the world strive to achieve this goal. IoT can modernize the existing power systems to achieve their global energy needs. IoT applications in power systems, smart grid monitors and managing power consumption, improving and monitoring wind turbines, photovoltaic installations, water pressure in radiation systems, radiation surface surveillance and the sounds of leakage alerts in nuclear power plants, and AC-DC power efficient controllers are all supplying the required energy with the least waste [10]. Intelligent device communications and high access of current power systems across all components such as generation, transmission, distribution and consumption, are all forming a flexible, consistent, secure and sustainable energy supply network [11].

A. Power Generation with IoT

The aim of power generation is to become inexpensive, accessible, sustainable to decrease the usage of fossil fuel and discharges. For this purpose, many generation companies around the world are progressively using the IoT. Here are some essential elements affecting power generation with IoT.

1) Remote Asset Monitoring/Management

The interconnected sensors are used for calculating the parameters to define the complete health of belongings from generating stations to transmission and distribution systems. The use of IoT in power generation also provides information about safety concerns such as gas emission before it leads to harmful situations for human life.

2) Process Optimization

IoT can be very helpful for generating stations by providing the actual information about the running conditions of power plants which is very useful for plant automation. Real time data is used to develop smooth operations for the plant, it increases fuel energy conversion and reduces maintenance costs.

3) Integration and Management of Renewable Energy Sources

The fundamental aim of a power plant is to eliminate fossil fuels and emissions by merging the different renewable energy sources such as, solar and wind, with coal and gas generating stations. IoT provides complete information about peak periods which helps to preserve sufficient energy to fulfill the demand for energy in peak periods.

B. Power Transmission and Distribution (T&D) with IoT

The problems in the transmission and distribution processes are somewhat alike. They contain fault detection, line losses, breakdown of insulators, breakage of conductors, etc. These issues can be monitored, controlled and solved through IoT. Smart meters, smart inverters, advanced distribution management systems (ADMS) and distribution network sensors are the common implemented examples of IoT devices in power T&D networks [12].

1) Asset Management and Maintenance

IoT is helpful to manage the equipment which usually is operated at substations by remotely monitoring with various types of sensors, that monitor parameters such as temperature, humidity and also detect the safety risks for insulators and pole structures before any unwanted event occurs.

2) Grid Assessment

IoT has the capability to give the actual information required to efficiently tackle crowds on T&D lines. To prevent unpredictability, the grid can certify that the connected generating stations have encountered the requirements of frequency-to-voltage control through IoT. In addition, IoT can help to decrease power loss and data theft during transmission and distribution using various sensors-based tracking features for the T&D lines.

3) Grid Contribution

One of the major contributions for grids is connecting IoT with renewable energy based generating plants with different levels, which will cause different changes in voltage across different nodes of the grid. The IoT leads these changes to address the flow of electricity by providing real time data and information. The solution is to automatically adjust the grid to sustain constancy.

4) Load Estimation

The deployment of sensors at diverse stations and alongside distribution lines can deliver real time information on power delivery in various areas, allowing the supplier organizations to make automatic and smart verdicts about voltage control, network configuration, load switching and more.

C. Power Consumption with IoT

In power system structures, consumption is the part where IoT has the greatest influence. The following are some of the key opportunities that IoT energy users can achieve [12].

1) Smart Decision Making

IoT is helping the consumers to make smart decisions according to their budget. For this purpose, smart meters are used to notify the consumers through a specific mobile phone application so they can save costs concerning their energy usage by adjusting their power consumption. Users

can disconnect the power supply to certain devices and set their desired conditions. With this advanced feature, they can eliminate waste and improve their consumption.

2) New Power Resolutions

With new business prototypes, there are also new IoT based energy solutions that assist monitoring, low productivity and energy storage for users. We are slowly approaching a future in which suppliers can deliver sustainable and affordable electricity to consumers and they can choose to buy energy periodically when tariffs fall, based on their usage.

3) Reduced Interruption

In developed countries a new line of smart meters capable of bilateral communication between distribution stations and users is being advanced. These meters send timely notifications and other important operating data to public service organizations. These organizations can process this data and reply more rapidly by fixing the errors and other issues. The meters also deliver real time statistics (load forecasts) that help the network modify the energy distribution as a result of peak time changes in various areas.

4) Power Sell to Grid

Enabling IoT technologies can benefit small households to sell unneeded energy from different sources such as wind plants and solar panels to the grid. With technology such as "Vehicle to Grid" even electric vehicles can start to supply the network with extra and unused energy.

5) Building a Zero Net Energy

IoT is also driving customer driven perceptions, such as zero net energy production. Zero net energy means that all the energy requirements of this home are primarily generated from renewable energy sources.

III. Emerging 5G Technology with IoT

A. Overview of 5G Technology

Communication is an important part of IoT that allows devices to communicate with each other along with services and applications. Data transmission between devices and applications should be safe and reliable. According to communication standards, various communication technologies can be classified with respect to wireless and wired communication systems [13], [14]. Wireless communication systems are currently preferred over wired for various reasons and over a variety of applications with reliable costs at lower speeds. This expands the infrastructure and provides easily accessible connections even in remote areas. There are various factors, such as operating costs, environmental issues and resource availability, which must be considered when choosing the right and stable communications system. Developing a new communications infrastructure with existing wireless technologies will be more beneficial in power systems where additional costs and time can be avoided [15]. This establishes an update on new wireless technologies for future communications systems with smart power networks [16]–[21].

For this purpose, 5G cellular communication networks are more multi-functional and flexible. This provides a larger platform to support many critical issues, and also solve many of the problems associated with cost analysis and energy consumption [22]. 5G networks are expected to set a larger scale deployment of the fourth industrial revolution. This ensures the convergence of widespread broadband, perception and intelligence, which causes greater changes in society. New wireless networks can bring the IoT to future power systems by providing greater benefits to utilities and consumers. Historically, among public networks in wireless modes, 5G has tended to develop a need for an energy paradigm. 5G networks can lead the new business models with leading edge and fog computing along with intelligent controls [23]–[25]. Fig. 1 shows the evolution of different wireless communication technologies by years.

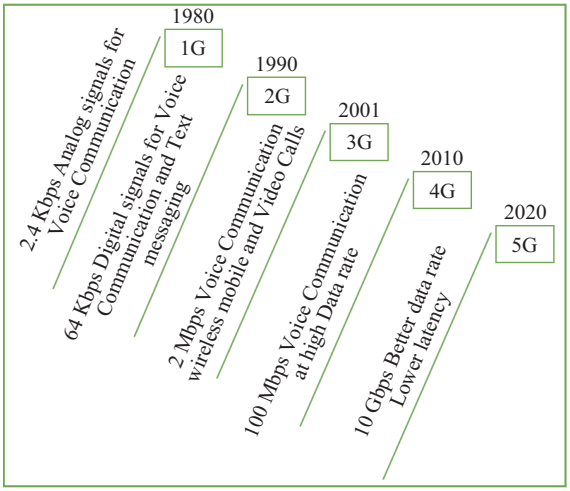

Fig. 1. Evolution of wireless cellular communication technologies by years.

B. 5G Cellular Technology

In advance cellular communications, 5G technology is the fifth-generation mobile communication network technology. This new technology provides diverse abilities and encourages full networking among countries globally. Its highest data transmission speed is hundreds of time faster than the existing 4G networks, also having features of low power consumption, low cost, wide range and efficient connections for better communications [26]–[29]. 5G is similar to infrared signal or Bluetooth though, it is one of the base stations that can manage more than 200,000 terminals with a data transmission speed of tens of Gb per second and covering a distance of several km.

IoT is a combination of cloud computing services with software defined networks (SDN) along with the emerging 5G technology to achieve better output with efficiencies [30], [31]. Binary coding is a constant technique used in past cellular communication systems with wide coverage and also reduced packet losses. In 5G networks, this technique can be used to enhance smart grid communications for achieving broader coverage areas [32].

C. Features of 5G in IoT

1) Quicker Response, Better Speeds and Higher Efficiency

In IoT, the devices need to connect via a high-speed internet with quicker response time for efficient and reliable communication. As compared with the existing 4G LTE cellular network speeds, 5G can provide better response and it can boost output, services, power, safety and security in different manufacturing industries such as, utilities, chemical and fertilizer, oil and gas, transportation and logistics [33].

2) Wide Range of Supporting Applications

In the near future, the 5G network will support wide ranges of applications by providing significant advantages including, IoT health care systems, IoT industrial systems, IoT power sectors and machine type communication (MTC) applications [33].

3) Connectivity of Heterogeneous Devices

Connectivity between many devices at any time and in any place can provide vast profits and permit mobile networks for IoT to be more fruitful, extremely accessible, economic and offer environmental-friendly resources [34]. Currently, many industries realized the benefits of wide area coverage between distributed and heterogeneous devices in emerging 5G networks which can be helpful to increase the productivity and high efficiency in IoT applications.

4) Supporting Heterogeneous Network Services

The existing 4G LTE technology is supporting heterogeneous network services but it still has some constraints in protocols and devices under its control. For better support of heterogeneous network services, it is essential for connecting and addressing various devices with 5G technology, because only this technology can provide a platform to create a large IoT. With 5G, we can probably comprehend computing capabilities being merged with wireless communications [33].

5) Low End-to-end Invisibility

Another important area of IoT is the internet of vehicles, which involves interconnections between various vehicles where reduction and jamming of traffic possibilities should be made easier. Intelligent and smart vehicular communications can be implemented by utilizing the ubiquitous availability [34]–[39]. 5G potentials have a much longer time of one millisecond, compared to about ten milliseconds in 4G technology and offer minimal delays in IoT-based applications, such as adding autonomous automobiles, activating remote sensors, etc. It is used for considering very large bandwidth and low latency for these promising 5G networks. The 5G network offers wide bandwidth, low latency and improved security, which also facilitates reliable real time health monitoring [33].

D. Comparison of Different Wireless Cellular Networks

Over the past few decades, a steady and gradual development of communication networks has been observed worldwide, starting from the first generation (1G) to the fourth generation (4G). Through these developments, the number of smart devices is increasing rapidly in everyday life. At this point, new applications and new business service providers in the wireless technology market are also expanding. Table I presents the era of wireless technology over the past few decades and its advantages.

E. Role of 5G in Power Systems

The 5G networks in power systems could provide a high-quality service to connect with the highest scalability. The state estimate provides a complete estimate of the system in which the status variable is a completely remote configuration environment on all nodes of the network. The visibility of traditional state estimation limits the high virtual complexity and communication delays, which are caused by dynamic and distributed electrical networks. The 5G network implementation is easily overlooked, as it allows for advanced and distributed status estimation projects [40].

Communication disorders can negatively affect demand response control. This requires advanced communication technology to allow controlled feedback between the user and the energy provider. This can be achieved by efficient planning of small 5G cells [41]. It requires a business structure with 5G and intelligent networks, which incorporates several features

TABLE I
THE ERA OF WIRELESS CELLULAR TECHNOLOGIES OVER THE PAST FEW DECADES

IoT System, Requirements	Wireless Cellular Networks				
	1G	2G	3G	4G	5G
Deployment	1980	1990	2001	2010	2019 or 2020
Data Type	Analog Data	Digital narrow band Data	Digital broadband Packet Data	Digital broadband high throughput packet Data	Not defined yet
Cost	Less Expensive	Less Expensive	Expensive	Expensive	More Expensive
Coverage Area	Small	Small	Large	Large	Small
Bandwidth	800 MHz	1900 MHz	2100 MHz	2600 MHz	
Data Rate	2.4 kbps	64kbps	High (3.1 Mbps)	High (100 Mbps)	Highest (10 Gbps)
Latency	Higher	High	Low	Low	Lower
Scalability	No	Yes	Yes	Yes	yes
Power Consumption	Higher	High	High	Low	Lower
Advantages	Mobility	Long-lasting battery adapter, Safe	Better Internet Experience	High data rate	Very low Latency, Better Data coverage, no droppage of calls
Disadvantages	Very poor spectral effiency and poor control	Demand rates are very low	Performance Failure for internet	Expensive because of more power consumption	–

and adapts to the space of the 5G cellular network [42]. These frameworks provide a foundation to the power system as well as having many advantages, such as bilateral energy trading, dynamic energy pricing and active wireless transmission spectra between the grid and the network.

The 5G technology is broadly used in power systems. Depending on its performance characteristics, it may incorporate the following typical business necessities: 1) Intelligent dispersed distribution automation; 2) Low-voltage power information collection, smart car charging station/pile, distribution of huge terminal access requirements for information collection type uplink services such as power supply access; 3) Power T&D line monitoring status, high definition return video calls and telephone, power system inspection by remote monitoring and emergency communication; 4) Millisecond accuracy load control; 5) Active distribution network differential ultra-high reliability and ultra-low latency required for protection of industrial control downlink services [33].

IV. Emerging 5G Technology with PIoT

A. PIoT

PIoT is a wireless sensor network (WSN) built into the industry using IoT technology. The wireless sensors in the system allow the exchange of information among various interconnected devices in the power systems [43]. The main goal of PIoT is to implement all the features of the power system to identify the interconnectedness of everything within the system, human-computer interaction with advanced and sophisticated information, and communication technologies, like artificial intelligence and mobile internet. It works as a smart ecosystem with full state-of-the-art information, effective data processing and suitable application structures [44].

In China, PIoT is a significant component of the intelligent internet plus smart energy network. The evolution from the idea of smart grids to the precise PIoT pattern is an expected growth of the basis for the interconnection of data and information. The aim of PIoT is to gain all data and information by reducing the fundamental physical connections [45].

The PIoT has excessive potential for applications and it will bring great advancement in the future. Nowadays, millions of station devices are connected to SGCC systems (including 450 million meters and tens of millions of protections and control devices). The devices associated with the SG-eIoT system is scheduled by SGCC, which will amount to two billion, and the whole PIoT will be the largest IoT ecosystem for access devices [33].

B. PIoT Architecture

Generally, the PIoT architecture consists of four-layers which are perception layer, network layer, platform layer and application layer.

1) The perception layer is the lowest layer of the PIoT architecture. It depends on broadly distributed sensors; the main concern of this layer is to gather valuable data from whole parts of the conventional grid system, including generation to consumption and also bring out the primary data integration and calculation.

2) The network layer associates data and real time information by building up a WSN in power systems at the perception layer, based on 5G and other advance wired or wireless communication techniques.

3) The platform layer is realizing the supervision of different types of PIoT network and data, as well as working systems. Application Programming Interface (API) provides the application development to explore the big data. The platform layer provides capability to process the big data in the network and prepare it for the further processing in the application layer.

4) To achieve the aim of steady and safe operation for the power grid, the application layer looks for building an intelligent, smart and unified power network. It is also known as the highest layer of PIoT [33], [45], [46].

As mentioned in the above section, it is a comprehensive path from the network layer to the platform layer and the application layer to the identity of these layers. As IoT slowly improves and PIoT is still in its initial growing phase, the researcher should place emphasis on the development of the perception layer and network layer of the PIoT. The platform and application layers can be established only after the foundation of these two layers is built.

C. 5G System Supporting the PIoT

In the power sector, IoT is a cohesive model for humans, machines and equipment used inside and outside the grid, built by implementing 5G and IoT [46]. In PIoT, smart grids are regarded as the most important framework. It works on the conventions of all components connected to the grid network, which are well monitored and controlled in each task. Extensive data is used and network connectivity is also widespread among integrations, with central information being shared centrally. Therefore, an appropriate communication system for the future smart grid is an area that transfers and distributes all power with the overall management system. The bilateral link between the transmission and distributed area requires a lot of data with proper channeling. Wireless communication is important in a reliable, efficient and intelligent way to improve the traditional power grid into the smart grid in IoT networks.

The role of data traffic distribution in smart grid networks can be divided into two parts [47]–[51]. The first is a home area network (HAN) which includes basic connections such as smart meters and sensors in communication between utilities

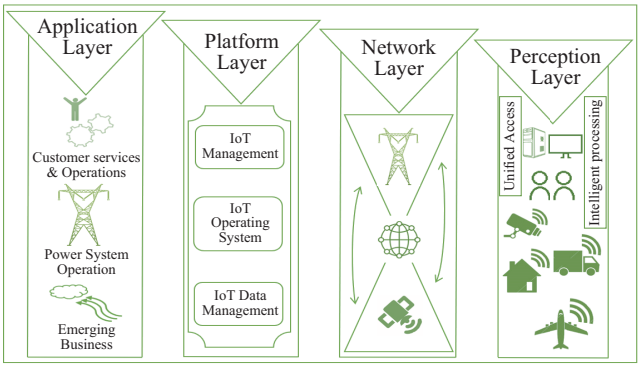

Fig. 2. PIoT Architecture.

and users [52]. The next important segment is the direct connection of public service providers and generation. Over the past decade, power line communication has been considered as the best communication provider between these segments. In the future, power line communication will allow the accuracy of the communication system and the integration of information [52], [53]. This is achieved by activating digital communication in addition to electrical power transmission.

Smart grid is a boom in PIoT, where the growth of productive and stable communication is constantly increasing. Therefore, bringing 5G to the smart grid makes it even more predictable to meeting growing energy needs. According to recent reports, by recognizing multiple power devices being connected to the power grid, 5G networks for power forecast monitoring are ready to extend the route for these connected devices and be able to connect immediately [54]. Therefore, connecting these devices with these 5G networks helps to effectively manage the energy balance and also helps to reduce energy costs. Effective data can be analyzed through a 5G network, which allows cities to execute their own energy projects according to population conditions. In addition, 5G networks help most distribution operators to achieve a monitoring level below the level of subsidies. This ensures adequate and balanced networking.

D. Challenges of Deployment of 5G in PIoT

Although 5G networks have great potential for use as a communication technology in an IoT network, these networks have not been completely fulfilling the requirements due to the following challenges. In a conventional network, due to the predictability of both the power distribution scheme for the end user and the generation system such as thermal and hydroelectric systems, it is much more manageable to plan the corresponding generation according to need.

1) Fluctuation

The main challenge faced by the PIoT network is the sudden change in a large number of small stations generating and fluctuating energy consumption at the end-user level. In addition, the traditional electric grid is very specialized towards the sending end and receiving end. In relation to the smart grid, the end user also becomes the producer. The communication techniques involved in the smart grid should be able to control the end user level. The key challenge in introducing any new technology is standardization.

2) Interoperability and Compatibility

The 5G network is expected to be standardized by 2020. New technologies must have interoperability and compatibility with existing technologies. The existing infrastructure of the power network must be utilized optimally and modifications must be made to make it compatible with the 5G standards. The basic requirements of each wireless network are not only the spectral efficiency but also the effectiveness of energy parameters as well. The emergence of 5G networks with PIoT will cause a multiple-fold increase in the amount of connected devices. This will lead to a significant increase in network power consumption. Thus, it is imperative to implement a sustainable network with renewable energy sources, optimal use of energy and minimal energy leakage.

3) Authentication

Authentication of network devices is very important on any network and requires a delay of hundreds of milliseconds on an existing system. This will be a problem in 5G networks, because it does not provide latency [55]. There is an exchange between delay and security with the privacy provided by the network.

4) Security

The threat of cyber-attacks is a serious challenge that must be maintained by communication networks. Survival during a power outage and black start is the main issue for any electricity network, because inaccessible electricity is not acceptable, even for a short time [56]–[59]. This requires technical solutions of 5G, which allows switching between different communication technologies, thereby ensuring uninterrupted delivery of information in real time.

5) Scalability

Scalability can be defined as the ability of a system to optimally manage growing needs [60]. For security reasons, PIoT with a robust infrastructure (which leads to the development of PIoT architecture), dispatch control and data collection by Supervisory Control and Data Acquisition (SCADA) provides the necessary adaptability and enhanced security [61]–[63]. Taking into account SCADA, the process includes three main blocks: the man-machine interface, the remote unit and the main terminal unit [64]. The SCADA system in the form of a network of stores offers more secure communication modules, which to a greater degree allow transmission sources. It also provides several security mechanisms for building secure key management using group communication. It provides secure localization and integrity and requires a secure level of communication at the highest level. The need for a transaction that transforms data proves that SCADA is a major concern for self-healing features. The PIoT network shows the entities required for high reliability to enable 5G to provide advanced communication between wireless networks, such as communication networks [61]. The main requirement for scalability is appropriate network planning to find and use the appropriate capacity necessary to meet future requirements [57]. In addition, this network should work autonomously when availability is limited in critical situations. To achieve this, numerous studies have been conducted, some of which relate to an integrated photovoltaic system [65].

E. Recommendation

An integrated component of an intelligent network using 5G is the exchange of information between electrical devices distributed in the PIoT network. The analysis of communication delay characteristics is necessary for designing a reliable network [66]. Recently, the mobile IoT federation as a service paradigm was proposed to improve the performance of 5G-based PIoT networks [67]. It includes a device pool managed by the IoT cloud provider. The need for an energy efficient communication model in recent years has attracted considerable attention from researchers. An important attribute of 5G in solving this problem is green communication [68]–[75]. To achieve this, it is necessary to determine the minimum

number of small cells to be active at different times in order to meet user needs and minimize power consumption.

V. Conclusion

The rapidly emerging concept of IoT is paving the way to achieve the goal in our modern life of combining several smart devices, applications and technologies. This article explores the fundamentals of this idea and provides modern research to control, manage and monitor the generation, transmission, distribution and consumption of power systems through IoT. Apart from the huge advantages of IoT for power systems, some associated challenges (such as, power management, security, reliability and big data) and their proposed solutions were presented in this article. Furthermore, this paper has reviewed the exceptional features of 5G cellular network technology emerging with IoT as the next generation network for supporting the new service requirements. More precisely, this article delivers detailed information on the role of the 5G cellular network in the progress of IoT and power systems. The concept and architecture of PIoT were discussed in this study. Finally, some of the problems associated with the deployment of 5G in PIoT were also presented in this study.

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